CLASS

ES 51 Turf Wars Competition

TEAM MEMBERS

Abby Yoon, Alex Ko, Fotis Paganis, Grant Kaufmann, Jack Kovac, Sarah Li

Seymur Galore Design Review 2

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MISSION STATEMENT

Problem: Design a compact, durable, and agile robot that can reliably collect and place designated objects on a central grid within a dynamic, ramped arena. The robot must fit within the constrained 11" cube startup size, operate under a 9A power limit, and navigate both 15° and 30° ramps. It must also be robust enough to survive multiple rounds and capable of recovering from failures using a limited number of lives.

Mission Statement: Our mission is to engineer a *lightweight*, *compact*, and *maneuverable* robot that excels in object manipulation, strategic placement, and reliable navigation under timed conditions. Through thoughtful design, precise fabrication, and iterative testing, we aim to maximize scoring efficiency and ensure operational resilience in a high-pressure, head-to-head tournament environment.

SPECIFICATIONS

Requirement	**Specification**	**Value**
Compactness	Starting Size	<11” x 11” x 11’
Usability	Setup and Breakdown Time	<2 minutes
Efficiency	Competing Time	<4 minutes per match
Power	Current Draw of Motors	9 A
Accuracy	System Success Rate	> 75% success with 5 robot lives
Mobility	Ramp Climbing Ability	15° and 30° ramps must be climbed
Manipulation	Object Handling Capability	=> 2 types of objects must be transported
Durability	Survive Match Damage	=> 3 matches should be completed without critical repair needed

PROJECT PROGRESS

PREVIOUS PHASES

CURRENT PHASE

PROJECT RESEARCH

Research competition requirements, document field parameters, create ideal strategy

BRAINSTORMING + PROTOTYPING

Brainstorm robot mechanisms, create cardboard prototypes, make CAD models

DESIGN SELECTION

Choose the best design using a systematic method of selection (Pugh Matrix)

FINAL DESIGN + MANUFACTURING

Complete design made, manufacture parts, assemble robot, test prior to competition

REVIEW OF STRATEGY

- Deposit two blocks first (in the center and corner)
- Go for hockey pucks next
- Score dog toys if possible but prioritize getting tic-tac-toe

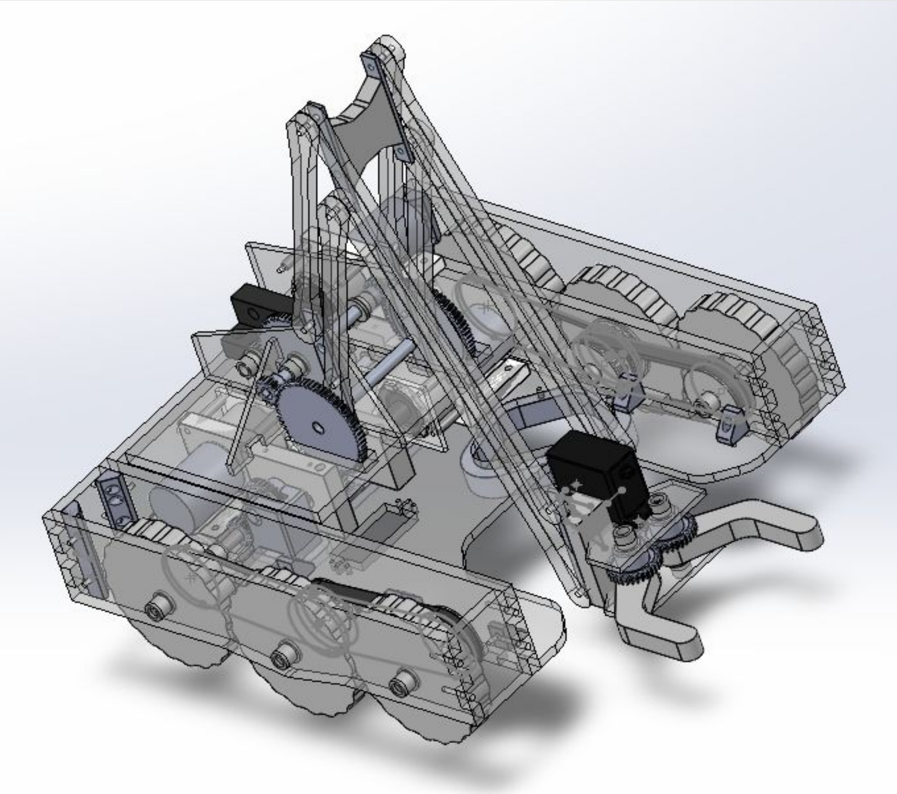
HIGH SCORING

Our strategy aims to win every round while scoring the greatest number of points. This will help us obtain a favorable ranking moving into playoffs.

ADAPTABILITY

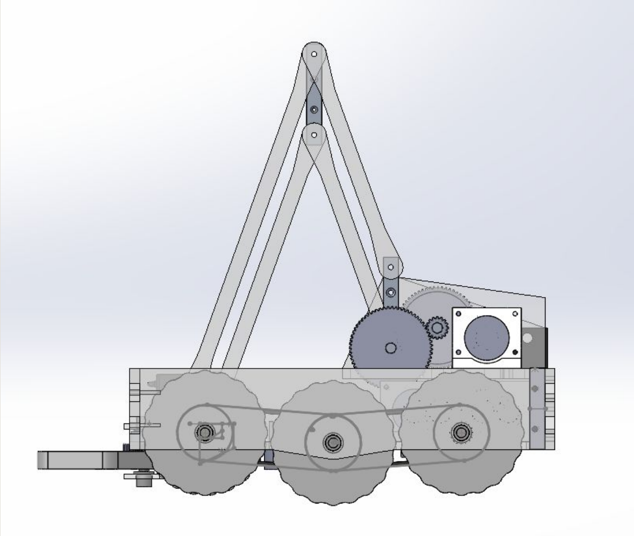
Our simple strategy allows for changes depending on our opponent. Sometimes we will try to win fast, other times we will try to deposit every scoring element.

Design Walk Through



Material Selection:

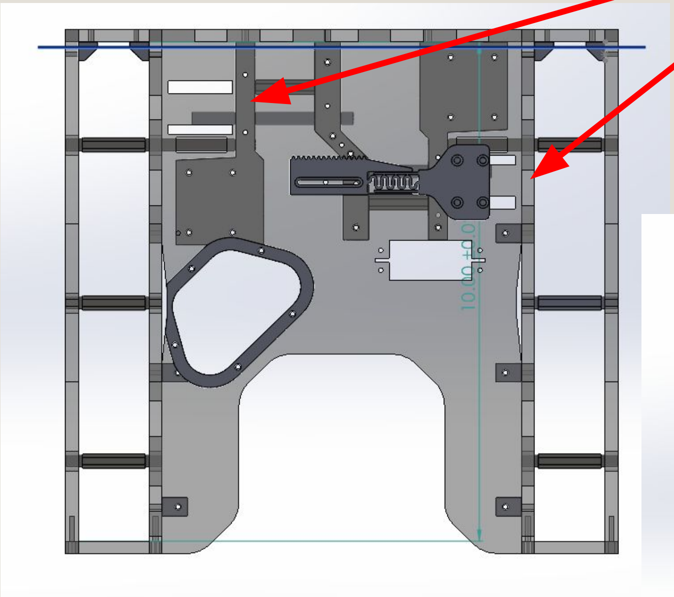
- Transparent = acrylic
- Metal shafts indicated
- All else 3D printed due to complexity of shapes and for efficient prototyping

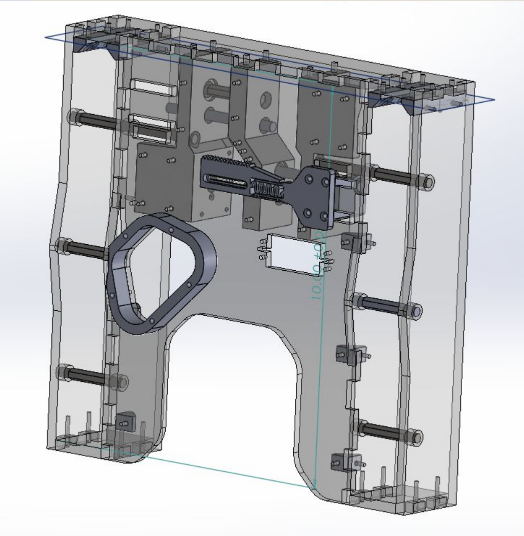


CHASSIS:

2 sliders

- Allows us to switch between 2 gear ratios using a servo motor
- GR 0.75 for traversing 30º incline
- GR 0.33 for traveling fast on ground plane





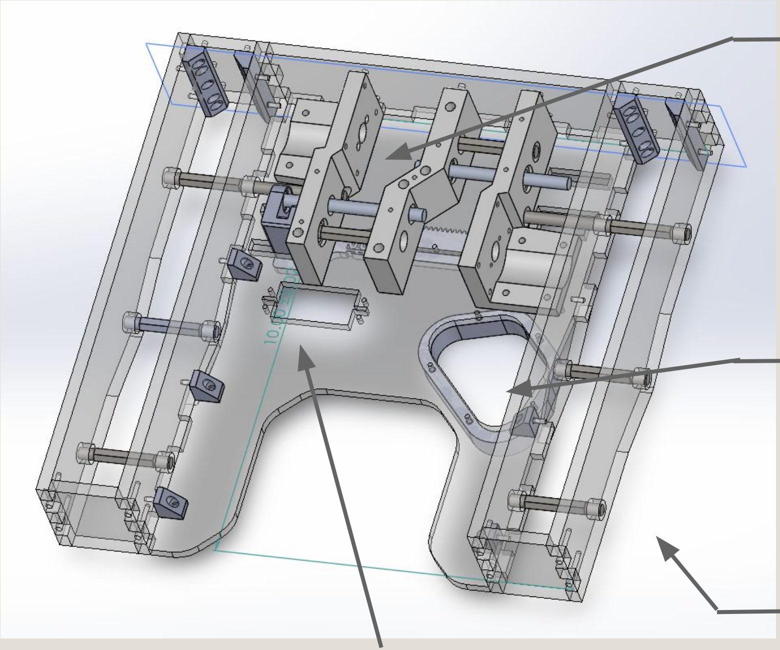
U-shaped:

- Allows for gripping mechanism

Wheels:

- Outer 4: Dragonskin (hard)
- Middle 2: (soft), mounted slightly lower than big wheels to reduce resistance during turns + improve balance
- Contrast in hardness → improved grip, smoother turns
- 10 teeth per wheel

CHASSIS:



Gears:

- 1:2 GR on output of gearbox
- Within gearbox it switches between a 3:2 ratio and a 2:3 ratio

Battery:

- Mounted close to center of robot to maintain stability

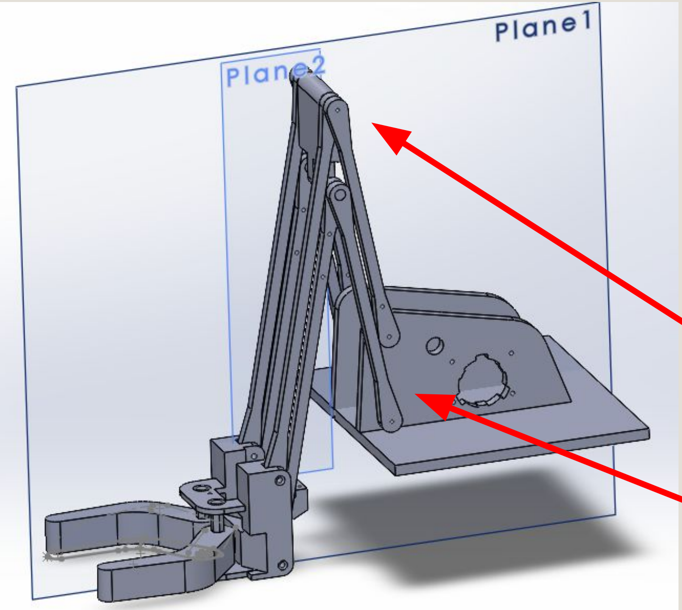
Supports 3 wheels rather than 2:

- More wheels = more contact with ground = better traction
- Distributes friction, allows turning
- Lowers center of gravity, enhancing overall stability

Servo motor controls sliders

- Gives us ability to use robot controller to change speed before and after inclines

FOUR BAR ARM



Arm thickness:

- Stiff enough to hold up under load

Arm height:

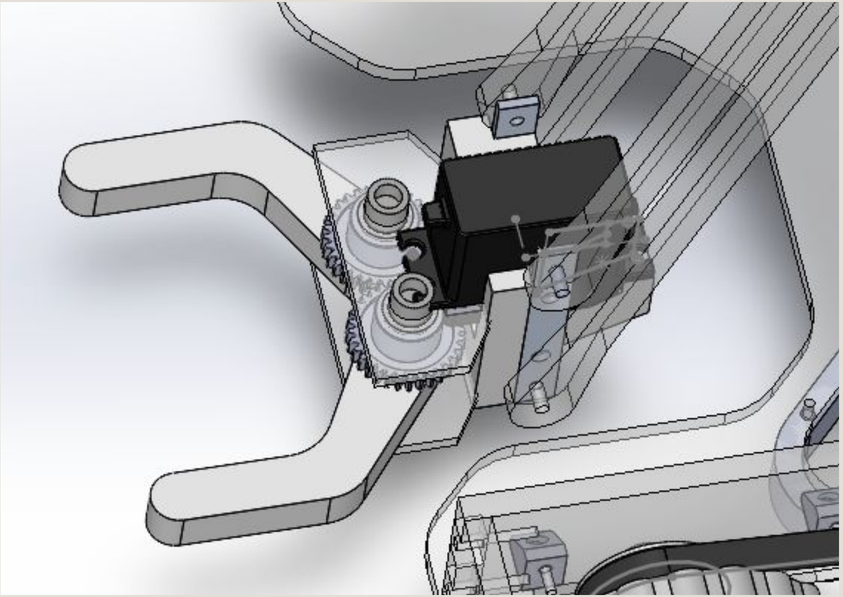
- Max under 11" constraint

Controlled by gear motor, which is powered by a battery

Degree of rotation: 80

Degree of rotation: 80

CLAW



Lightweight

Extends backward, allowing robot to fit inside Box of Justice

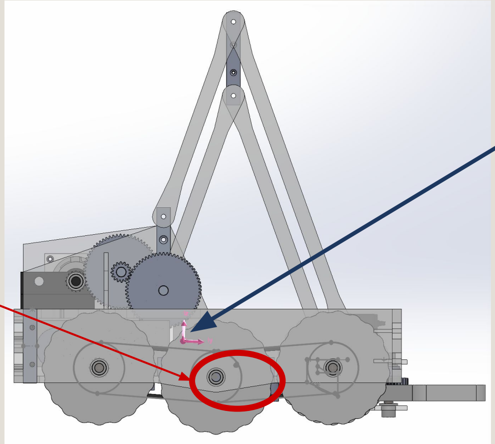
Movement/rotation powered by servo motor

Criteria	Baseline	Weight	Design 1	Design 2	Design 3	Final
Full Height	0	1	-1	0	+1	+1
Design Simplicity	0	2	+1	-1	0	-1
Circuit Simplicity	0	1	+1	-1	0	0
Control Simplicity	0	3	0	-1	+1	+1
Intake Capability	0	3	0	+1	-1	+1
Drop Off Mechanism	0	2	-1	+1	0	+1
Style	0	much!	-1	+1	0	+1
Total	-	-	-1	0	1	8

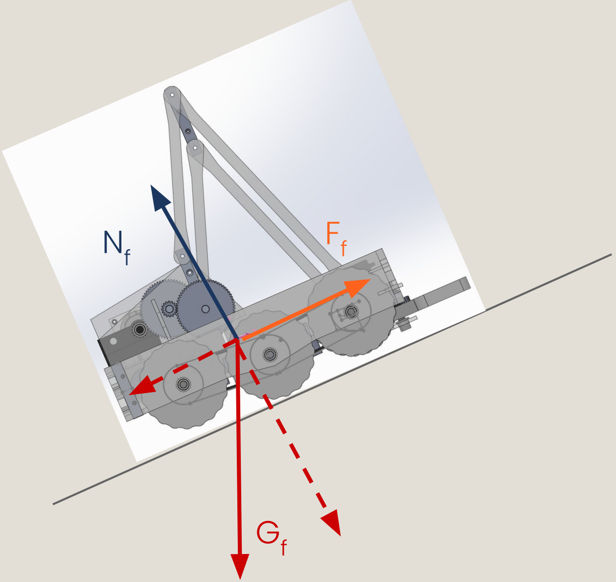
Physics Overview

COM AND RAMPS

Approximate battery location



COM



Center of Mass (Without Battery)
0.99 inches behind and 0.89 inches above the center axel, 0.1 inches off centerline

Drive Train Calcs:			
Mass(kg):	3		
Wheel Dia(M):	0.075		
@Max Incline	Fx = mgsin(30)	14.7	N
Axle Torque:	0.55125	NM	
Needed Ratio:	0.7986741219	(2 MOTORS)	
Linear Speed:	1.090934417		

$G_f = 9.81(3) = 29.43$ N
$N_f = \cos(30)*G_f = 25.49$ N
$F_f = \sin(30)*G_f = 14.71$ N

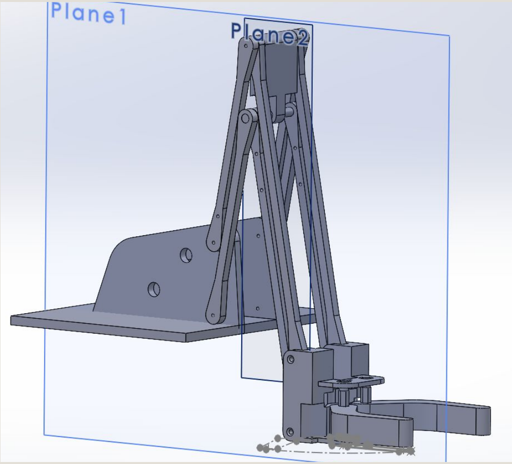
The center of gravity sits over the point in the axel plain 0.99 + sin(30)*0.89 = 1.43 inches behind the center axel
The axels are ~3.2 inches apart
Thus the center of gravity sits ~3.2-1.43 = ~1.77 inches in front of the real axel, meaning the robot is still stable

ARM TORQUE REQ.

- Complex mechanism, easiest to calculate for M(dz/dt) and solve for the torque from there

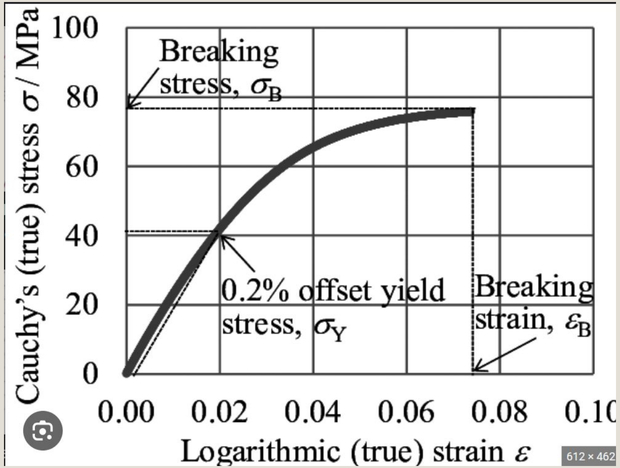
- Still settling on an exact gear ratio

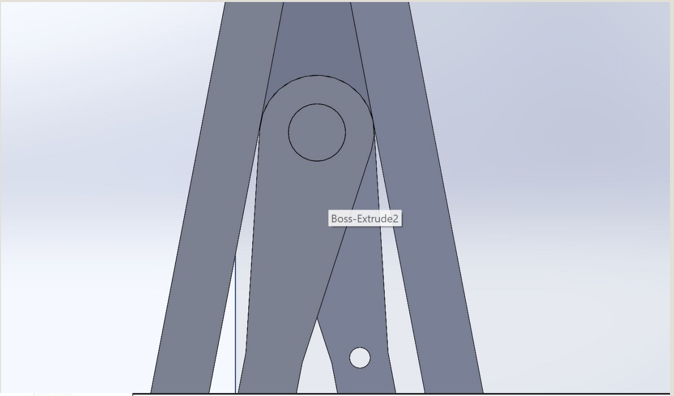
Arm Motor T:		
Max Arm Torque	0.369189	NM



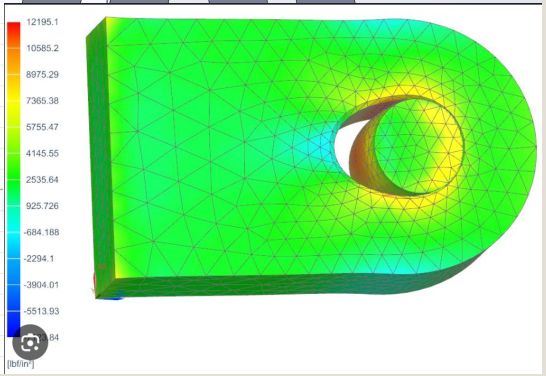
CRITICAL FORCES

- While we originally considered AL, we can use acrylic with a FS of about 100





	In/Lb			Kg/M
Arm Length	17	Arm Length	0.4318	
Arm Mass	0.38	Arm Mass	0.171	
Payload Mass	1	Payload Mass	0.153	
Gripper Mass	0.3	Gripper Mass	0.135	
Beam Width	0.375	Beam Width	0.009525	
Beam Thickness	0.25	Beam Thickness	0.00635	
Beam Separatioi	2	Beam Separatioi	0.0508	
Pin Diameter	0.0625	Pin Diameter	0.0015875	
		Stress:	36549.9473	Pa
			0.0365499473	MPA
		Joint Bearing Stress:		
			219299.6838	Pa
			0.2192996838	MPA



Manufacturing Progress

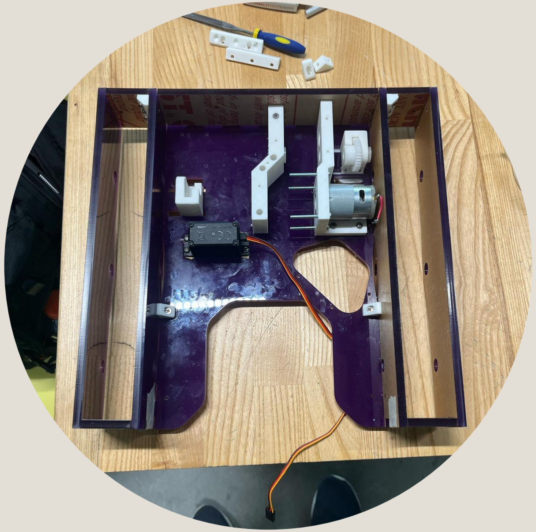
MANUFACTURED PARTS

1. WHEELS

Wheel molds were cut on the CNC machines. Wheels were made using Dragon Skin silicone with an acrylic insert for stability.

2. CHASSIS HOUSING

An acrylic box-like housing that balances structural rigidity, easy open access, and a lightweight frame.



3. AXLES

Manufactured from hex shafts using the lathe.

MANUFACTURING NEXT STEPS

01

COMPLETE CHASSIS ASSEMBLY

- Manufacture final parts (wheels, shafts)
- Assemble chassis
- Test chassis on ramp

02

FOUR BAR ARM

- Need to cut acrylic plates and assemble
- Will likely make a prototype to test before

03

CLAW

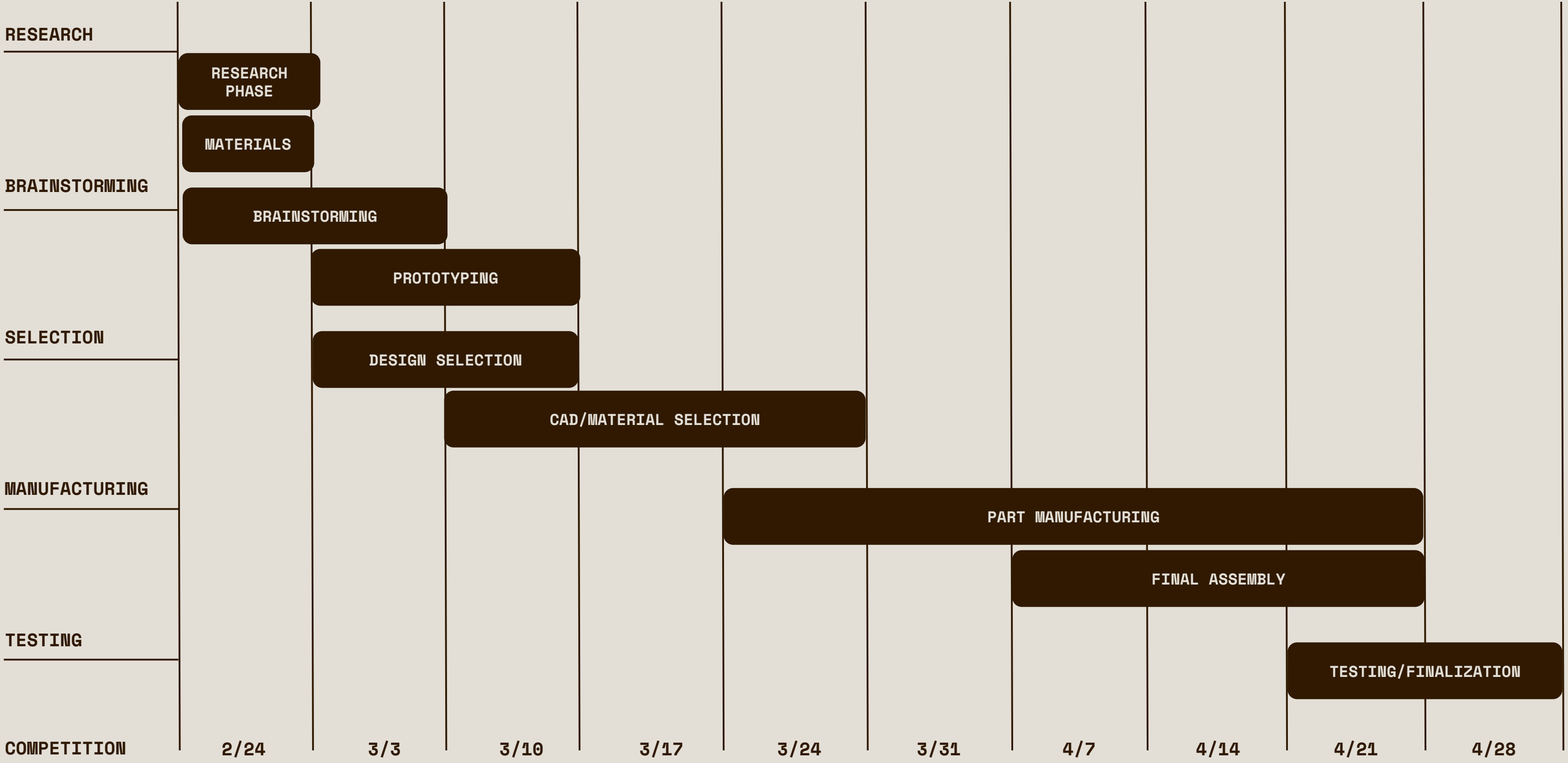
- Needs to be prototyped
- Potential minor redesign coming
- Manufacturing will likely require silicone and acrylic

04

TESTING/COMPETITION

- This will occur multiple times throughout the process (chassis, arm, on-field run throughs)

Gantt Chart


RESEARCH
RESEARCH PHASE
MATERIALS
BRAINSTORMING
BRAINSTORMING
PROTOTYPING
SELECTION
DESIGN SELECTION
CAD/MATERIAL SELECTION
MANUFACTURING
PART MANUFACTURING
FINAL ASSEMBLY
TESTING
TESTING/FINALIZATION
COMPETITION
2/24
3/3
3/10
3/17
3/24
3/31
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4/14
4/21
4/28

Thank You

SEYMUR GALORE 04.10.25